NUMERICAL INVESTIGATION OF AMMONIA–DIESEL SPRAY AND MIXING BEHAVIOUR IN COMPRESSION IGNITION ENGINES

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Background :

1. Compression ignition engines are widely used because of their **high efficiency and power density**.
2. Conventional diesel combustion is associated with **CO_2 and pollutant emissions** and dependence on petroleum-derived fuel.
3. **Ammonia** is a carbon-free fuel that can reduce the carbon intensity of engine operation.
4. Ammonia–diesel dual-fuel combustion combines:
5. **Ammonia as the main carbon-free fuel**
6. **Diesel as a pilot/ignition source**
7. Spray development and fuel–air mixing strongly influence the subsequent combustion process.

Therefore, understanding ammonia–diesel spray and mixture formation is an important step before analysing combustion.

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Importance of Spray and Mixing :

Importance of Spray and Mixture Formation

1. Fuel Injection
2. Atomization & Breakup
3. Droplet Motion
4. Evaporation
5. Air Entrainment
6. Fuel–Air Mixing
7. Combustion

Key spray characteristics

1. Spray penetration
2. Droplet size and velocity
3. Spray dispersion
4. Evaporation
5. Fuel–air distribution

Spray development determines the spatial and temporal distribution of fuel available for combustion.

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AMMONIA–DIESEL SPRAY PROBLEM :

Ammonia	Diesel
Carbon-free fuel	High-reactivity fuel
Different thermophysical properties	Established CI fuel
Different evaporation behaviour	Well-established spray behaviour
Requires dedicated spray investigation	Reference for spray modelling

Different physical and injection characteristics can influence spray development, evaporation, interaction and mixture formation.

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LITERATURE REVIEW :

1. Diesel Spray Fundamentals

- Atomization Spray penetration
- and breakup
- Entrainment and mixing

2. Numerical Spray Modelling

- Eulerian–Lagrangian approach
- Lagrangian parcels
- KH/RT breakup models
- Droplet forces and evaporation

3. Ammonia Spray

- High-pressure liquid ammonia sprays
- Distinct spray/evaporation behaviour

4. Ammonia–Diesel Spray Interaction

- Spray interaction
- Injection configuration
- Mixture formation

5. Engine-Level Studies

- Combustion characteristics
- Performance and emissions
- Numerical + experimental investigations

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RESEARCH GAP :

What has been studied?

- Extensive investigation of conventional diesel spray development.
- Established numerical approaches for:
  - Lagrangian spray modelling
  - Breakup and atomization
  - Droplet motion
  - Evaporation
- Recent studies on:
  - Liquid ammonia sprays
  - Ammonia–diesel spray interaction
  - Ammonia–diesel engine combustion and emissions

What requires further investigation?

- Spatial and temporal distribution of ammonia and diesel during spray development.
- Interaction between the two sprays and its effect on mixture formation.
- Spray/mixing behaviour under **CI-engine-relevant conditions.**
- Influence of selected injection and operating parameters.
- A numerical framework that can be systematically extended from spray modelling towards combustion analysis.

Focused numerical investigation of ammonia–diesel spray and mixture formation is required as a foundation for subsequent combustion and emissions analysis.

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OBJECTIVES :

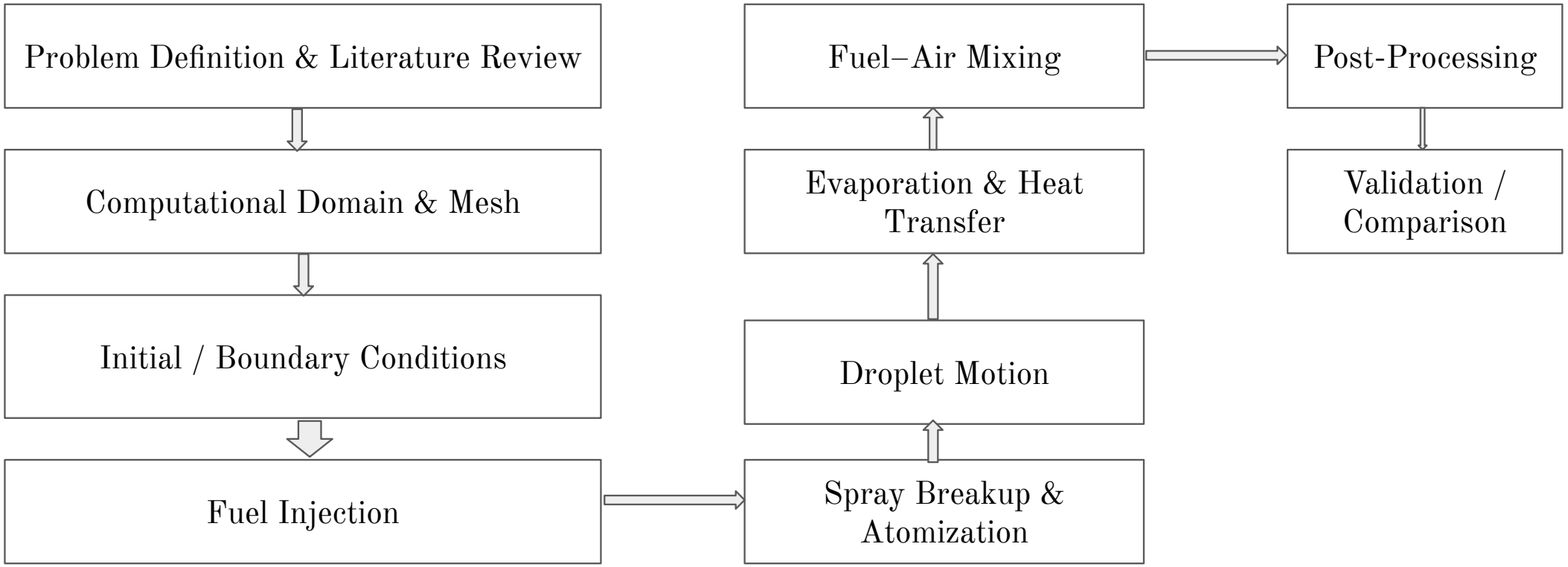
Main Objective : To numerically investigate ammonia–diesel spray and mixing behaviour under CI-engine-relevant conditions.

Specific Objectives :

1. Investigate fundamental liquid spray development and penetration.
2. Develop a numerical framework for fuel injection, breakup, droplet motion and evaporation.
3. Analyse ammonia and diesel spray/mixing characteristics.
4. Investigate spatial and temporal fuel distribution during mixture formation.
5. Study the influence of selected injection and operating parameters.
6. Establish a foundation for subsequent combustion and emissions analysis.

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METHODOLOGY :



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PRELIMINARY NUMERICAL INVESTIGATION :

Initial verification of the Lagrangian liquid-parcel framework

Computational framework

1. **Software:** OpenFOAM-14
2. **Solver:** multicomponentFluid
3. **Approach:** Lagrangian parcel tracking
4. **Case:** Single liquid water parcel in a gaseous domain

Purpose

The preliminary simulation was performed to verify:

1. Lagrangian parcel tracking
2. Droplet motion
3. Heat transfer
4. Phase change / evaporation

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COMPUTATIONAL SETUP :

Parameter	Value
Computational domain	$0.1 \times 0.1 \times 0.01$ m
Number of cells	125
Parcel diameter	100 μm
Initial parcel mass	5.23599×10^{-10} kg
Initial parcel velocity	0 m/s
Initial parcel temperature	350 K
Simulation time	0.5 s

Models used

1. Gravity
2. Droplet drag
3. Ranz–Marshall heat transfer
4. Liquid evaporation

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PRELIMINARY RESULTS :

Parameter	Initial	Final
Liquid Mass	5.23599×10^{-10} kg	3.29855×10^{-11} kg
Particle Temperature	350K	≈319.73 K

Successful Lagrangian parcel tracking and evaporation were observed.

The preliminary simulation successfully verifies the basic liquid-parcel tracking and evaporation framework.

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PROPOSED FUTURE WORK :

Proposed further steps:

1. Diesel Spray Simulation
2. Breakup & Evaporation
3. Validation with Published/Experimental Data
4. Ammonia Spray Simulation
5. Ammonia–Diesel Spray Interaction
6. Mixing & Mixture-Formation Analysis
7. Parametric Investigation
   a. Injection pressure
   b. Injection timing
   c. Fuel ratio
   d. Operating conditions
8. Extension towards Combustion & Emissions

The immediate next step is to establish and analyse a transient diesel spray case as the reference configuration.

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CONCLUSION / CURRENT STATUS :

Literature

1. Diesel spray modelling is well established.
2. Ammonia–diesel systems are receiving increasing attention.
3. Focused spray/mixing investigation remains relevant.

Present work

1. OpenFOAM-14 Lagrangian framework implemented.
2. Preliminary liquid-parcel case successfully simulated.
3. Parcel tracking, heat transfer and evaporation observed.

Next stage

1. Diesel spray → ammonia spray → ammonia–diesel interaction → mixing analysis.

Current status: Numerical framework verification completed; transition to fuel-spray simulations is the next stage.

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Thank You!